

Course Outline

INTRODUCTION TO AR, VR AND MIX REALITY

Course Code: **BAVM34601**

Semester VI

B. Des (Hons)

Communication Design 2023-27

School of Arts & Design

3 Credits

Prepared by	Navdeep Sindhu, Dr. Vijay
Email ID	navdeep.sindhu@wosen.edu.in

Brief Description and Relevance of the Course Title: INTRODUCTION TO AR, VR AND MIX REALITY

This cutting-edge course introduces communication design students to the transformative potential of Augmented Reality (AR), Virtual Reality (VR), and Mixed Reality (MR) in shaping future user experiences. Focused on **project-based learning and hands-on activities**, the course enables students to explore immersive technologies through real-world design challenges. By working with tools like Unity, Blender, and AR/VR headsets, students will gain practical skills in spatial design, interaction design, and immersive storytelling. Emphasis is placed on designing human-centered, multi-sensory experiences that blend the digital and physical worlds, preparing students to innovate at the forefront of communication design.

Objectives

1. Understand the fundamental concepts and theories of AR and VR technologies.
2. Explore the design principles and best practices for creating engaging and interactive AR, VR, MR experiences.
3. Develop the technical skills to create basic AR and VR and MR applications using relevant

Program LOs Addressed:

PLO	Definition
PLO1: Visual Communication	Demonstrate ability to create effective visual communication through various media, including print, digital, and interactive formats.
PLO2: Brand Identity Development	Develop strong brand identities through visual language and messaging.
PLO3: User Experience (UX) Design	Understand user-centered design principles and create user-friendly interfaces and experiences.
PLO4: Research and Analysis	Conduct thorough research to inform design decisions and identify target audiences.
PLO5 : Storytelling and Narrative	Effectively communicate ideas and messages through visual storytelling and narrative design.

Course LOs

CLO	Mapping to PLO
CLO1. Students will demonstrate a thorough understanding of augmented reality (AR), virtual reality (VR), and mixed reality (MR) technologies, including their underlying principles, applications, and potential impact on various industries.	PLO1, PLO3, PLO4
CLO2. Students will develop proficiency in designing immersive experiences for AR, VR, and MR platforms, applying principles of user interaction, interface design, and immersive storytelling to create compelling and user-centered experiences.	PLO1, PLO3, PLO5

CLO3. Students will acquire practical skills in developing AR, VR, and MR applications, using industry-standard development tools and platforms to prototype, iterate, and refine immersive experiences.	PLO1, PLO3
CLO4. Students will be able to critically evaluate AR, VR, and MR experiences, analyzing their effectiveness in engaging users, conveying information, and achieving desired objectives, and making informed design decisions based on their analysis.	PLO3, PLO4

Prerequisites: NIL

Reference/ Reading Material Recommended

Code	Textbook/Article/Report Name	Edition/Year/Link	CLO Mapped
R1	Augmented Reality: Principles and Practice by Dieter Schmalstieg and Tobias Hollerer, 1st Edition.	First Edition, 2012	1, 3 & 4
R2	Learning Virtual Reality: Developing Immersive Experiences and Applications for Desktop, Web, and Mobile by Tony Parisi, 1st Edition.	Second Edition, 2015	1 & 2
R3	Virtual Reality by Steven M. LaValle, 1st Edition.	First Edition, 2020	1 & 2
R4	The VR Book: Human-Centered Design for Virtual Reality by Jason Jerald, 1st Edition.	First Edition, 2015	2 & 3
R5	Practical Augmented Reality: A Guide to the Technologies, Applications, and Human Factors for AR and VR by Steve Aukstakalnis, 1st Edition.	First Edition, 2016	3 & 4
R6	Virtual Reality Insider: Guidebook for the VR Industry by Sky Nite	First Edition, 2014	2 & 3
R7	Mixed Reality in Architecture, Design, and Construction by Panos Mavros	First Edition, 2009	4

Session-Wise Topics and Reading/References

Sn.	Topic	Session Intended Learning Outcome	Pedagogy	CILO	Reading Material Code
1	Introduction to Unity and AR Development	- Overview of course objectives and expectations - Introduction to AR/VR/MR (Extended Reality) - Introduction to Unity Interface - Understanding Unity's components: GameObjects, Assets, Scenes, Prefabs - Basic Navigation and Scene Setup.	Lecture	1	R1
2	Introduction to AR in Unity	- Introduction to Augmented Reality (AR) concepts - Overview of AR Foundation and XR Plugin - Setting up an AR project in Unity	Lecture, Activity & Discussion (Project Based Learning & Collaborative workspaces, encouraging peer interaction)	1 & 2	R2 & R3
3	Image tracking in AR	- Introduction to Image Trackers - Implementing Image Tracking using AR Foundation and XR plugin - Creating and importing Image Targets - Simple interaction with Image Targets (e.g., display 3D model upon detection) Assignment/Exercise (A1): - Develop an AR app which shows 3D model based on Image Tracking.	Lecture & Discussion (Collaborative workspaces, encouraging peer interaction)	1 & 2	R2 & R3

4	Plane Tracking with AR	• Introduction to Plane Tracking • Implementing Plane Tracking	Activity (Project Based Learning)	1 & 2	R2 & R3
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		using AR Foundation • Placing objects on detected planes • Understanding plane detection boundaries Assignment/Exercise (A2): Develop an AR app which shows 3D model based on Plane Tracking.			
5	AR App Development – Adding Interactions	• Adding user interactions with AR objects (e.g., tap to rotate, scale) • Implementing UI elements for AR interaction (Logo) • Combining Image and Plane Tracking for complex AR apps	Lecture & Activity (Project Based Learning)	2	R5
6	Advanced Unity AR & Introduction to Unity VR	• Advanced AR Features • Customizing AR experiences with scripting • Adding environmental effects to AR scenes • Optimizing AR apps for different devices	Lecture	2	R5
7	Introduction to Virtual Reality (VR) in Unity	• Introduction to Virtual Reality concepts • Setting up a VR project in Unity • Overview of supported VR platforms (Oculus, HTC Vive, etc.)	Activity (Project Based Learning)	1 & 2	R5
8	VR Interaction and Environment Design	• Creating immersive environments in VR • Designing and importing 3D models for VR • Understanding VR navigation and interaction basics.	Lecture	3	R6

9	MR Interactions, Final Project Presentations and Feedback	• Adding interactions in VR (e.g., locomotion, object manipulation) • Scripting interactions for VR experiences. • Presentation of completed AR/VR/Twinmotion projects • Peer and instructor feedback • Course wrap-up and final discussions on future applications of learned skills. Assignment/Exercise (A3): • Develop an MR app which shows 3D model and add basic interactions with model.	Activity (Project Based Learning)	3	R6
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Performance Evaluation Components for the Course

Session No.	Marks	Evaluation Form	CLO
Continuous Evaluation/Internal Assessment			
1 - 5	15	Conceptual Clarity & Comprehension Skill Exploration Basics of Design Principles	1,2
6 & 8	15	Research and Comprehension	1,2
9-12	30	Reflective and Critical Thinking Quality of Outcome and Documentation	2,3 & 4
End Term Jury Evaluation			
After Course Completion	40	Final Project Presentation and Feedback	4

AI Assessment

Session No.	Marks	Evaluation Form	AI Assessment Level	Scale Name	CLO	Outcome Measured
1 - 5	15	Conceptual Clarity & Skill Exploration Understanding of XR concepts, Unity environment, and basic AR features	No AI	1	1	Outcome Measured
6 & 8	15	Research & Comprehension Exploration of AR/VR precedents, toolsets, and scripting	No AI	1	1	Outcome Measured
9 - 12	30	Reflective & Critical Thinking + Output Quality Design thinking, interaction testing, peer feedback, and documentation	No AI	1	2,3 & 4	Outcome Measured
End Of Semester	40	Final Project Presentation, Execution & Feedback	No AI	1	3 & 4	Outcome Measured

Evaluation Components & CLO Alignment

The evaluation structure for the **INTRODUCTION TO AR, VR AND MIX REALITY** course is designed to build student competencies in a progressive and structured manner—from foundational ideation (such as concept development, scripting, and storyboarding) to full-cycle production planning, prototyping, and execution. This phased approach ensures **holistic learning**, allowing students to evolve from basic XR understanding to the creation of fully functional immersive experiences.

Evaluation Component	CLO Alignment	Description
Conceptual Clarity & Skill Exploration (Component 1)	CLO 1	Tests students' ability to develop an original story and write a structured short film script.
Research & Comprehension (Component 2)	CLO 1	Assesses visual planning, sequence design, and storytelling through image-based layouts.
Reflective & Critical Thinking + Output Quality (Component 3)	CLO2,3 & 4	Demonstrates team collaboration, visual execution, and narrative synthesis.

CLO1. Explain the core principles, functionalities, and limitations of AR and VR technologies.

CLO2. Analyze and evaluate various design considerations for developing effective AR/VR experiences.

CLO3. Demonstrate proficiency in using relevant software tools to create basic AR experiences.

CLO4. Demonstrate proficiency in using relevant software tools to create basic VR experiences.

Evaluation Rubrics

CLO 1: Explain the core principles, functionalities, and limitations of AR and VR technologies.

Criteria	Exceeds Expectations	Meets Expectations	Does Not Meet Expectations
Marking Band	>80	55-79	<55
Understanding of AR/VR Principles	Deep and accurate understanding; explains concepts with clarity and examples.	Good understanding ; explains key concepts with some clarity.	Limited or unclear understanding; vague or incorrect.
Technical Functionalities	Clearly explains tracking, rendering, and interaction systems with examples.	Basic explanation of core functionalities.	Incomplete or inaccurate explanation.
Limitations & Constraints	Identifies and critically explains key technical and user-related limitations.	Lists common limitations with basic reasoning.	Misses or misrepresents key limitations.
Real-World Application	Effectively applies concepts to relevant scenarios.	Applies concepts to general examples.	Little to no application to real-world contexts.

CLO2. Analyze and evaluate various design considerations for developing effective AR/VR experiences.

Criteria	Exceeds Expectations	Meets Expectations	Does Not Meet Expectations
Marking Band	>80	55-79	<55
UX Design	Intuitive, immersive, and user-focused.	Functional with basic user engagement.	Confusing or lacks user focus.

Interaction Design	Effective, responsive, enhances experience.	Works but limited in depth.	Inappropriate or poorly implemented.
Design Justification	Clear, well-reasoned decisions.	Basic rationale provided.	Weak or no justification.
Usability	Thoughtful and inclusive.	Meets standard usability needs.	Lacks usability awareness.

CLO3. Demonstrate proficiency in using relevant software tools to create basic AR experiences.

Criteria	Exceeds Expectations	Meets Expectations	Below Expectations
Marking Band	>80	55-79	<55
Software Use (Unity, AR Foundation, etc.)	Confident, error-free use of tools with advanced features.	Adequate use with minor errors or limited features.	Struggles with basic tool use or setup.
AR Implementation	AR experience functions smoothly with clear user interaction.	Basic AR functionality implemented correctly.	AR setup is incomplete or not functional.
Problem Solving & Debugging	Identifies and resolves issues independently and efficiently.	Solves basic issues with some guidance.	Requires constant help; unresolved issues remain.
Output Quality	Polished, creative, and technically sound AR prototype.	Functional and meets basic requirements.	Lacks functionality or clarity.

CLO 4: Exploring - Collaboratively plan, shoot, edit, and present a product video showcasing their chosen design.

Criteria	Exceeds Expectations	Meets Expectations	Below Expectations
Marking Band	>80	55-79	<55
Software Use (Unity, XR Toolkit, etc.)	Uses tools effectively with advanced VR setup and features.	Uses basic tools correctly with minor issues.	Incomplete or incorrect tool usage.
VR Environment Creation	Well-structured, immersive environment with functional interactions.	Basic environment with limited but working interactions.	Poorly designed or non-functional environment.
Navigation & Interaction	Smooth, intuitive navigation and responsive interactions.	Standard navigation and basic interactivity.	Clunky or ineffective interaction setup.
Technical Execution	Optimized, bug-free VR experience.	Mostly functional with minor bugs.	Unstable or non-working VR prototype.

Attendance & Punctuality

Learning is an interactive process. Students are expected to be present in all the classes. Absence is only appropriate in exceptional circumstances. Voluntary activities are never valid reasons for missing any class.

Students may refer to the student handbook for regulations covering attendance.

Students who do not meet attendance requirements will not be permitted to write the end term examination and will be required to repeat the course with the next batch of students.

Late arrival is disruptive to the learning environment; students are expected to be in class before the scheduled commencement time. Students arriving for class after the scheduled commencement time will be turned away unless they have a valid reason to be permitted to attend.

Copyright

The content provided by the faculty in the class is copy-righted. Students are instructed not to distribute or share content used during courses with external entities.

Student Code of Ethics

Each student enrolled in this course accepts personal responsibility to uphold and defend academic integrity and to promote an atmosphere in which all individuals may flourish. The Students' Code of Ethics strives to set a standard of honest behaviour that reflects well on students and the school. All students enrolled in these courses are expected to follow the Students' Code of Ethics contained in the student handbook. Unethical and unfair practices adopted by students may lead to penalties such as having to repeat the course or having the student's enrollment cancelled.

Appendix A: AR App -Image Tracking

Objective:

Create a basic AR application using Unity and AR Foundation that displays a 3D model when a specific image is detected.

Instructions:

1. Choose or design a 2D image to act as the tracking target.
2. Set up an AR project in Unity using AR Foundation.
3. Implement image tracking to display a 3D model when the image is detected.
4. Test the AR app on a supported device.
5. Document the process with screenshots and a short summary.
6. Submit the Unity project and a brief report in PDF format.

Deliverables:

- Unity project folder (zipped)
- PDF report with:
 - Project title & description
 - Screenshots of working AR app
 - Summary of process and tools used
 - Key challenges and solutions

Criteria	Marks	Description
Application	7	Proper setup and implementation of image tracking.
Skills Exploration	7	Creativity in scene design and optional interactivity.
Outcome & Documentation	8	Functional AR app and clear, well-documented report.

Appendix B: AR App -Plane Tracking

Objective:

Build an AR application using Unity and AR Foundation that detects horizontal or vertical surfaces (planes) and places a 3D model on them in real-world space.

Instructions:

1. Set up a Unity AR project using AR Foundation and XR Plugin.
2. Enable **plane detection** (horizontal/vertical).
3. Configure the app to place a 3D model on a detected plane when tapped or triggered.
4. Ensure the model stays anchored to the plane in real-time.
5. Test the app on a compatible device.
6. Submit the Unity project along with a brief PDF report.

Deliverables:

- Zipped Unity project folder
- PDF report including:
 - Project title & description
 - Screenshots of the app in action
 - Summary of tools, methods, and key learnings
 - Challenges faced and how they were resolved

Criteria	Marks	Description
Application	7	Accurate implementation of plane tracking and object placement.
Skills Exploration	7	Use of features like interaction (e.g., tap to place, scale, rotate).
Outcome & Documentation	8	Functional app and clear, well-structured documentation.

Appendix C: MR App -3D Model with Interaction

Objective:

Develop a Mixed Reality (MR) application using Unity that displays a 3D model and allows users to interact with it using basic input gestures (e.g., tap, drag, scale, rotate).

Instructions:

1. Set up a Unity project using AR Foundation or XR Toolkit, suitable for MR deployment.
2. Display a 3D model anchored to a real-world surface or fixed in mixed space.
3. Implement **basic interactions** like tap to animate, scale, rotate, or move the object.
4. Ensure smooth interaction feedback and intuitive control.
5. Test on a supported MR device or simulate interactions within Unity.
6. Submit the Unity project along with a concise PDF report.

Deliverables:

- Zipped Unity project folder
- PDF report including:
 - Project title & concept
 - Description of interactions implemented
 - Screenshots of MR experience
 - Tools used, methods applied, and brief reflection

Criteria	Weightage	Description
Application	8	Correct setup of MR environment and object placement.
Skills Exploration	8	Implementation of meaningful and functional interactions.
Outcome & Documentation	14	Quality of interaction, user experience, and clarity of documentation.

Appendix D: End Term Jury Assessment

Overview:

This final assessment evaluates students on the culmination of their learning in the **INTRODUCTION TO AR, VR AND MIX REALITY** course. Through a jury-based review, students will present their final immersive project—demonstrating conceptual clarity, technical execution, design process, and professional readiness. Evaluation will be based not just on the final output but also on the journey, decision-making, and ability to communicate the concept effectively.

Instructions:

1. **Develop and present your final immersive media project**, integrating AR, VR, or MR features along with completed assignments (A1, A2, A3).
2. **Deliver a professional presentation** to the jury, clearly explaining your:
 - Idea and concept
 - Design and technical approach
 - Execution process
 - Final experience/outcome
3. **Reflect** on your personal and project growth, challenges faced, and key insights gained during the course.
4. **Dress professionally** and be prepared to confidently answer questions from the jury regarding your work, tools used, and design rationale.

Assessment Focus Areas:

- Conceptual Depth & Originality
- Design Process & Technical Execution
- Interaction & User Experience Design
- Clarity of Communication & Presentation
- Professional Conduct & Readiness

Evaluation Criteria for Jury Assessment

Criteria	Weightage	Description
Product	16	Final outcome: quality, creativity, completeness, and alignment with the intent.
Process	16	Depth of ideation, exploration, iteration, and application of Filmmaking principles throughout.
Presentation	4	Clarity, confidence, structure of delivery, and ability to articulate the concept and choices made.
Personality	4	Professionalism, preparedness, ownership of work, engagement with jury feedback, and overall attitude.

Course Code: BAVM34601	INTRODUCTION TO AR, VR AND MIX REALITY					
		L	T	P	O	Total Credits
		1	1	2	0	3
Objectives						
1. Understand the fundamental concepts and theories of AR and VR technologies.						
2. Explore the design principles and best practices for creating engaging and interactive AR, VR, MR experiences.						
3. Develop the technical skills to create basic AR and VR and MR applications using relevant						
Outcomes						
CLO1. Explain the core principles, functionalities, and limitations of AR and VR technologies.						
CLO2. Analyze and evaluate various design considerations for developing effective AR/VR experiences.						
CLO3. Demonstrate proficiency in using relevant software tools to create basic AR experiences.						
CLO4. Demonstrate proficiency in using relevant software tools to create basic VR experiences.						
Reference Materials						
Textbooks (T) per availability in library						
Reference Books (R)	R1	Augmented Human: How Technology Is Shaping the New Reality — Helen Papagiannis				
	R2	Understanding Virtual Reality: Interface, Application, and Design — William R. Sherman & Alan B. Craig				
	R3	Learning Virtual Reality: Developing Immersive Experiences and Applications for Desktop, Web, and Mobile — Tony Parisi				
	R4	The VR Book: Human-Centered Design for Virtual Reality — Jason Jerald				
	R5	Designing Immersive 3D Experiences: A Guide for Museums, Brands, and Explorers — Renée Foss				
	R6	Reality Media: Augmented and Virtual Reality — Jay David Bolter, Maria Engberg, and Blair MacIntyre				
	R7	New Realities in Audio: A Practical Guide for VR, AR, MR and 360 Video — Stephan Schütze				
Journal Articles (J) provide link	J1	"Augmented reality and virtual reality displays: emerging technologies and future perspectives" A comprehensive, open-access review of AR/VR optical display technologies, challenges, and future directions. irjstem.com+2nature.com+2link.springer.com+2				

only		
Case Material/ Whitepaper provide link only		1. "Virtual, mixed, and augmented reality: a systematic review for immersive systems research" A detailed systems-oriented review covering AR, VR, and MR technologies with application insights. link.springer.com [PDF Available] 2. "Augmented Reality: A Review of Applications" (via

		Academia.edu) A general survey of AR use cases, technical background, and concept differentiation from VR. academia.edu [PDF Available] 3. "Beyond the Screen: Reshaping the Workplace with Virtual and Augmented Reality" (arXiv preprint) Discusses the XR (AR/VR/MR) continuum, technology evolution, and workplace applications. tandfonline.com+5arxiv.org+5link.springer.com+5 [PDF Available]
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Any Other Source s	Coursera – Intro to AR/VR/MR/XR (University of Michigan / Google AR & VR) Offers foundational training in AR, VR, MR concepts and development with Unity & ARCore en.wikipedia.org+11coursera.org+11programmingvalley.com+11en.wikipedia.org+2gamedevacademy.org+2coursera.org+2 FutureLearn – Introduction to Virtual, Augmented and Mixed Reality (Lancaster University) Covers XR fundamentals, applications, ethics, and emerging tools circuitstream.extendedlearning.ubc.ca+2classcentral.com+2coursera.org+2futurelearn.com+1classcentral.com+1 Class Central – Free VR & AR Course Listings Curated list of free courses from top platforms like Coursera, edX, and Udacity circuitstream.extendedlearning.ubc.ca+2classcentral.com+2classcentral.com+2 Additional Tools & Resources OpenSpace3D – A free, open-source no-code immersive XR engine for quick prototyping en.wikipedia.org Wikipedia – List of AR Software & SDKs – Discover tools like Vuforia, Spark AR, WebAR Studio, Lens Studio, etc. en.wikipedia.org Recommendation Guide If you're new to AR/VR/MR:
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		Start with FutureLearn or Udemy tutorials for basic theory and context. • For hands-on Unity development: Audit the University of Michigan courses or follow Zenva's tutorial guides. • To explore AR/VR design: Take the User Experience & Interaction Design specialization. • To prototype quickly: Experiment with OpenSpace3D for no-code XR development.
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Modules			
Module No.	Topics	Total Contact Hours	CLO Mapping
1	Introduction to Interactive Media • Introduction to AR and VR technologies • History and evolution of AR/VR • Applications of AR/VR across various industries	10	1
2	AR and VR Fundamentals • AR/VR hardware and software overview • Marker-based and marker less AR/VR experiences • AR/VR development tools and platforms • Hands-on workshops on creating basic AR experiences	15	2
3	MR Fundamentals • MR hardware and software overview • MR interaction methods and user experience design • 116 • Storytelling and content creation for MR • Hands-on workshops on creating basic MR experiences	15	3 & 4
4	Project Development • Identifying and defining AR/VR/MR project ideas • Project planning, design, and development • User testing and feedback integration • Project presentation and evaluation	20	3 & 4
Total Contact Hours		60	
Guest lecture / Expert sessions:			
Sl.No	Name of Resource Person	Module/Topics	Total Hours
Recommended by Board of studies		DD/MM/YYYY	
Approved by Academic Council on		No:	DD/MM/YYYY

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Disclaimer: The Syllabus and the Course Outline is indicative of the body of knowledge a student should possess. It does not anyway be construed as a to be covered in class in as-is condition. Faculty may add, delete and modify content based on updated information or context.

